Interpretable Machine Learning

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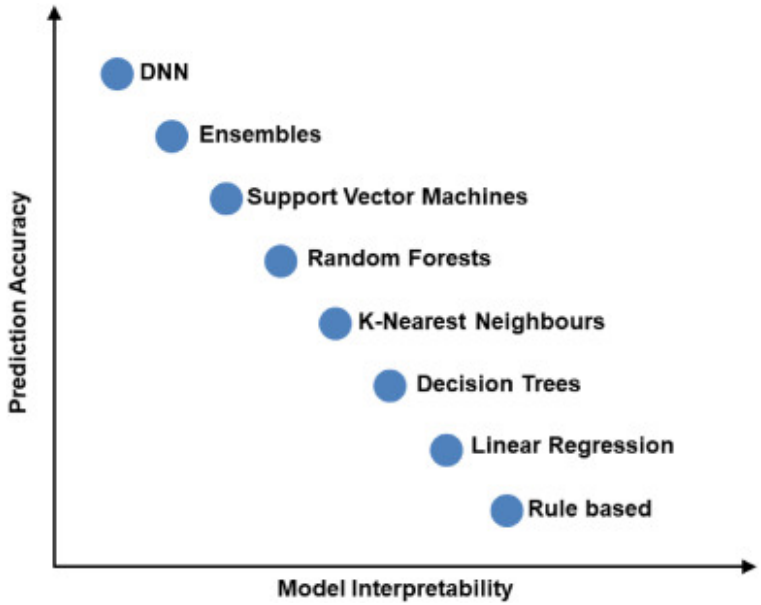


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Black- versus glass-boxes

I think that our XAI courses are popular because people want to "understand" what a black-box is doing.



Source: "Glass-box versus black-box machine learning models for clinical predictions: model transparency is not a universal solution", Molina, D., et al. (2023)

Black- versus glass-boxes

- Well-established notion of an interpretability-versus-accuracy trade-off
- Which is, of course, not universally true: sometimes, a glass-box model is as accurate as a black-box

This talk

- This is possible more often than you think
- Why does it matter?
- How to make it happen?
- Test-set performance is not all that matters

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Having it both ways

FICO challenge

- Explainable model for the risk of defaulting on a loan
- Organized on the premise of the model being a black-box
- Yet a "custom" interpretable model did best

Classifier	Test Accuracy	Test AUC
Random Forest	0.697±0.017	0.757±0.017
Boosted trees	0.723±0.024	0.789±0.028
SVM (linear kernel)	0.720±0.029	0.795±0.027
SVM (RBF kernel)	0.727±0.023	0.799±0.022
8-layer neural network	0.722±0.022	0.792±0.026
Logistic regression	0.731±0.023	0.801±0.028
2-layer additive risk model	0.738±0.020	0.806±0.025

Source: Rudin, C., Zhong, C., Semenova, L., Seltzer, M., Parr, R., Liu, J., ... & Boner, Z. (2024).

Logistic regression, decision tree and an additive model all did well; the tree, however, wasn't trained with a classic, greedy algorithm

Having it both ways

FICO challenge



Source: Chen, C., Lin, K., Rudin, C., Shaposhnik, Y., Wang, S., & wang, T. (2018). An interpretable model with globally consistent explanations for credit risk

Having it both ways

Pneumonia risk

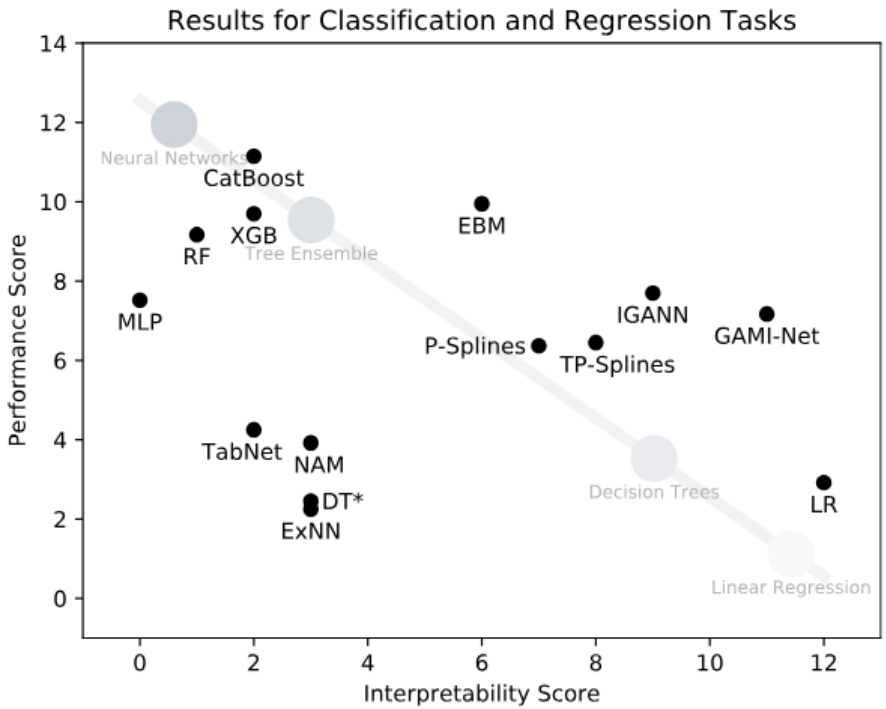
Predict the probability of death (POD) for patients with pneumonia so that high-risk patients could be admitted to the hospital while low-risk patients be treated as outpatients.

Model	Pneumonia
Logistic Regression	0.8432
GAM	0.8542
GA^2M	0.8576
Random Forests	0.8460
LogitBoost	0.8493

Caruana et al. (2015)

- 14,199 pneumonia patients, 1542 (10.86%) patients died from pneumonia; 46 features
- Note: logistic regression actually non-linear due to feature discretization

Having it both ways



Source: Kruschel, S., Hambauer, N., Weinzierl, S., Zilker, S., Kraus, M., & Zschech, P. (2025)

Even the simplest glass-boxes often account for a large fraction of the predictive power of the best-performing model.

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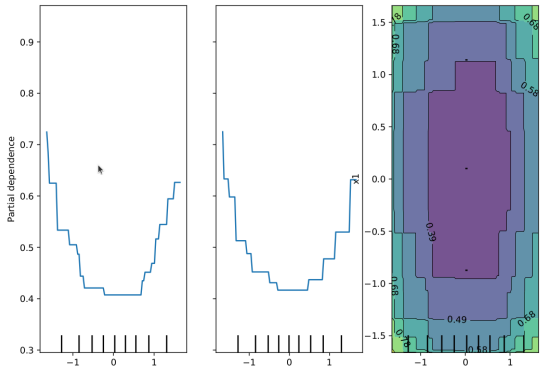
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Explanations $\neq$ model

Unlike for an interpretable model, which provides 'its own explanation', we explain a black-box via something else: variable importance measures, partial dependence plots, a surrogate model, etc.

The explanation of a black-box

- Not completely faithful to the model; if it was, we could use it instead of the black-box.
- Can be misleading when inconsistent with the model.



Explanations $\neq$ model

Why is this bad?

- "All models are wrong, and some are harmful"
- High-stakes decisions: Healthcare, justice system, credit scoring, ...
- Generalization, robustness, fairness: perhaps the model is picking up undesirable patterns from the data and you aren't aware.

Open-ended evaluation

- Validation sets can incentivize undesirable solutions, if data problematic.
- Interpretability gives us additional ways of evaluating the systems.
  "... interpretable models do not necessarily create or enable trust – they could also enable distrust. They simply allow users to decide whether to trust the model."

Data issues

For data that are abundant, unconfounded, complete, clean, iid, lack fairness issues, ...

Data issues

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For data that are abundant, unconfounded, complete, clean, iid, lack fairness issues, . . . , . . . black-box methods work great.

Data issues

For data that are abundant, unconfounded, complete, clean, iid, lack fairness issues, . . . , . . . black-box methods work great.

Data issues

- Confounding factors
- Selection bias
- Socially harmful biases
- Scarcity
- Wrong labels
- . . .

Real-world data

- Real-world data, especially tabular, are often marred by these issues
- For such data, relatively simple glass-box methods tend to be effective
- We may have domain knowledge that we can use to "correct" or compensate for some of the issues

Confounding

Pneumonia risk

- Large project in the 90's: predict the probability of death, for patients with pneumonia, so that high-risk patients could be admitted to the hospital.
- Run a clinical trial to determine if the machine learning prediction of risk, prior to hospitalization, can drive the decision about hospitalization.
- Neural nets clearly outperformed glass-boxes (AUC=0.86 versus 0.77 for logistic regression).
- Yet, they were considered **too risky for use on real patients and logistic regression was used instead**.

Confounding

Pneumonia risk

- A rule-based system, learned the rule "HasAsthma(x) $\Rightarrow$ LowerRisk(x)": patients with pneumonia who have a history of asthma have lower risk of dying from pneumonia than the general population.
- A true pattern in the data: patients with a history of asthma were usually admitted directly to the ICU, and the aggressive care was so effective that it lowered their risk of dying from pneumonia compared to the general population.
- Because the prognosis for these patients is better than average, models trained on the data incorrectly learn that asthma lowers risk, when in fact asthmatics have much higher risk (if not hospitalized).

Confounding

Pneumonia risk

- If the rule-based system had learned that asthma lowers risk, certainly the neural nets had learned it, too.
- The rule-based system was intelligible and modular, making it easy to recognize and remove dangerous rules like the asthma one.

Decision: **not use the neural nets**.

- The asthma problem could be solved, yet the **lack of intelligibility made it difficult to know what other problems might also need fixing**.
- Perhaps pregnant women with pneumonia also receive aggressive treatment that lowers their risk below that of the general population. The neural net might learn that pregnancy lowers risk, and thus recommend not admitting pregnant women, thus putting them at increased risk.

Confounding

More examples

- A neural network picking up on the word "portable" within an x-ray image, representing the type of x-ray equipment rather than the medical content of the image
- "Cow on the beach"
- Zip code as proxy for race or income
- ...

"Clever Hans" effect: The classifier works "well", but is arriving at conclusions looking at the wrong features.

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Glass-box models

- Roughly: A predictive model that is constrained so that a human can better understand its reasoning process.
- Provide their own explanations, which are faithful to what the model actually computes.

Alternative terms

- (Inherently) interpretable
- Transparent
- Intelligible

What makes a model interpretable?

- Small enough to inspect or reason about.
- Sparse and modular: parts can be checked independently.
- Inputs and transformations have a domain meaning.
- Useful constraints can be expressed: monotonicity, additivity, causality, ...

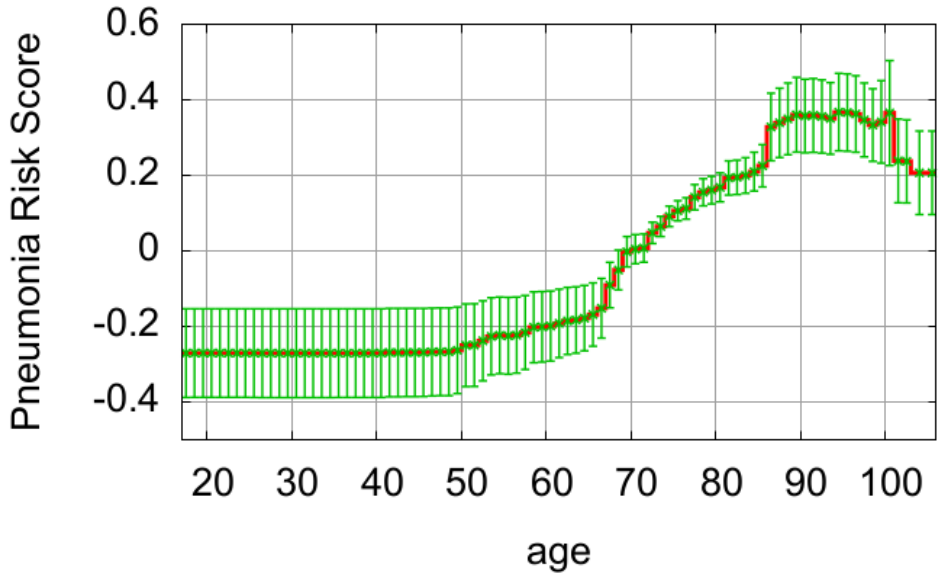
Glass-box models

Model families

- Logical: trees, rule lists/sets, ...
- Additive (+ interactions): generalized additive models, naive Bayes, ...
- Linear (+ interactions): logistic/linear regression, linear discriminant analysis, ..., RuleFit, ...
- Constrained neural-nets
- Hybrids: e.g., a shallow tree with additive models at the leaves

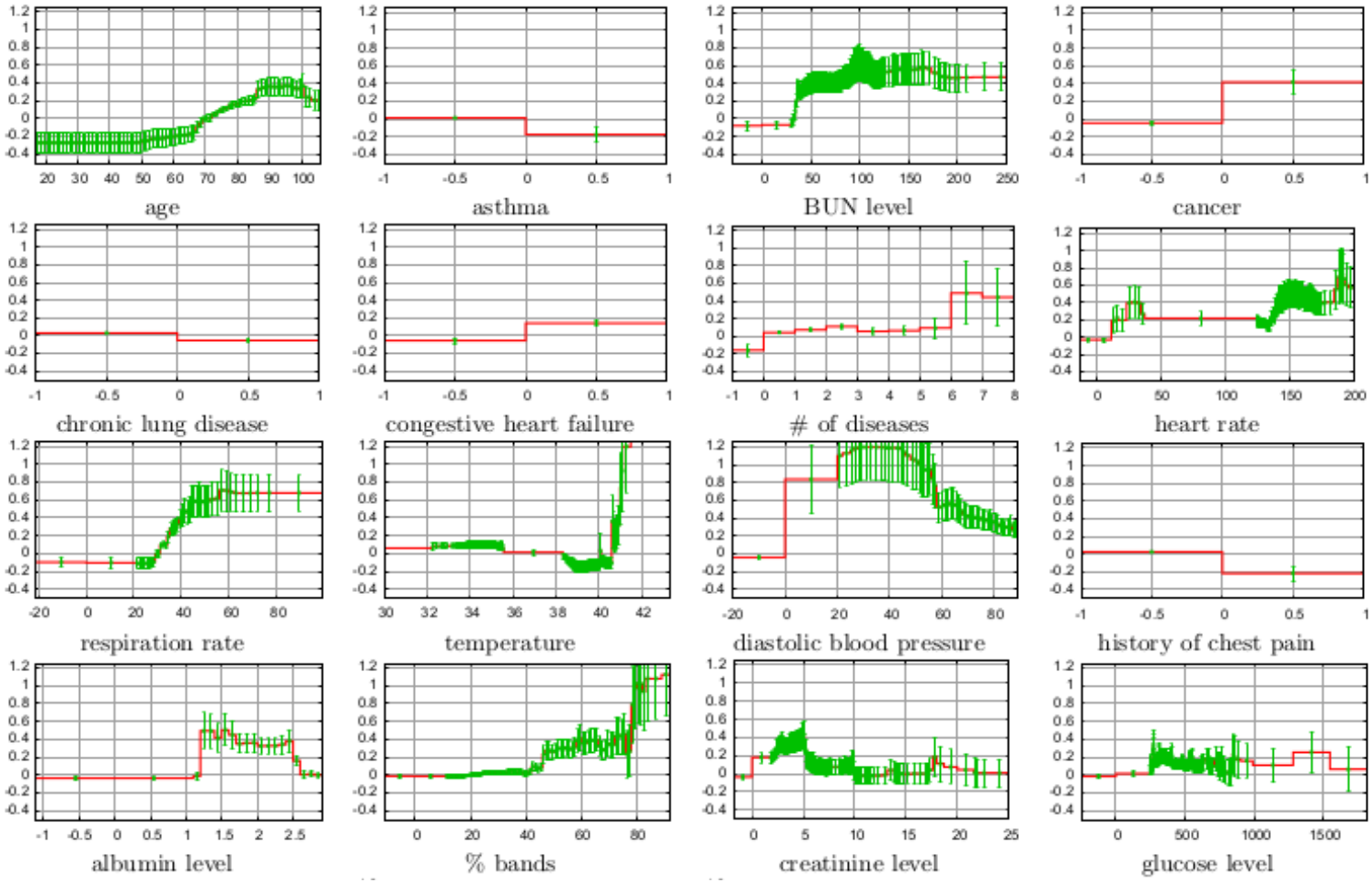
GAMs

A GAM prediction is built from feature-specific curves. Each curve shows how one feature moves a prediction above or below a baseline.



- Terms with scores above (below) zero increase (decrease) total risk over the baseline.
- Each term is interpreted with respect to the baseline.
- 80-year-olds are at more risk than the general population; 20-year-olds at less risk.

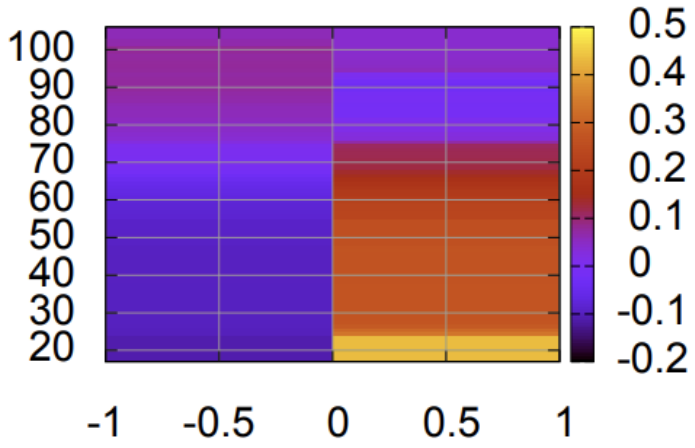
GAMs



- Each graph is centered vertically: mean prediction over training set is 0.
- Same Y-axis scale on all graphs for ease of comparison.

Interactions

Additive models can include a small number of interactions when a single-feature view is not enough.



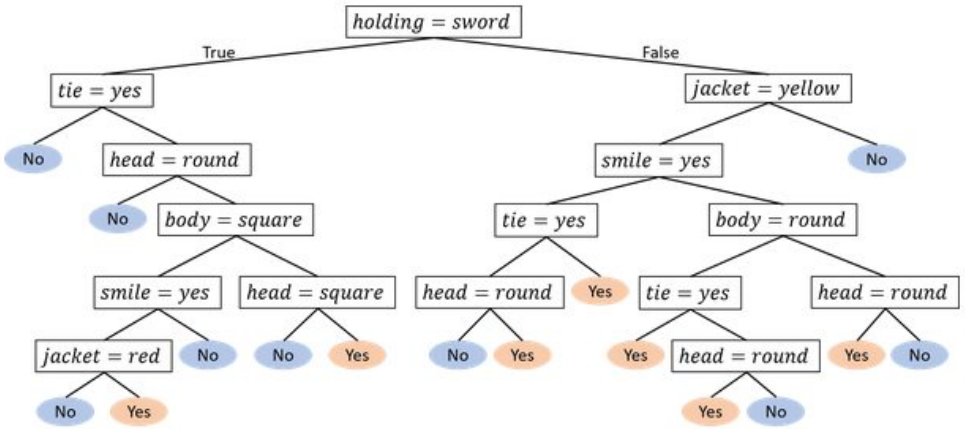
- Risk highest for youngest patients: probably cancers acquired in childhood and not cured by age 18.
- Interaction: risk declines with age for patients with cancer, but rises with age for patients without cancer.

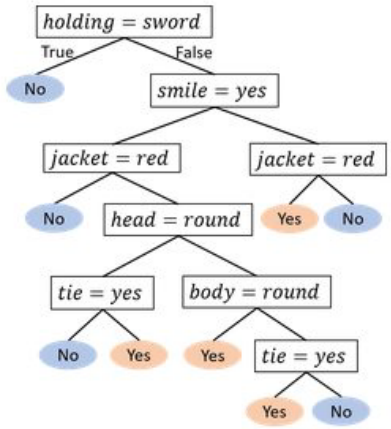
Multiple training algorithms

- There may exist an accurate and interpretable tree for your dataset, but CART can't find it
- That is: do not discard the representation (e.g., tree) because of the algorithm (e.g., CART)

Globally Optimal Sparse Decision Trees

Monk 2: Six features





Left

- 16-leaf decision tree learned by CART.
- Training accuracy: 75.74%; test accuracy: 69.44%.

Right

- 9-leaf decision tree generated by GOSDT.
- Training accuracy: 81.07%; test accuracy: 73.15.

Linear models

Ways to train a linear classifier

- (Bayesian) logistic regression
- SVM with linear kernel
- Linear discriminant analysis
- Lasso / ridge / elastic-net
- ...

Iterative training / preprocessing

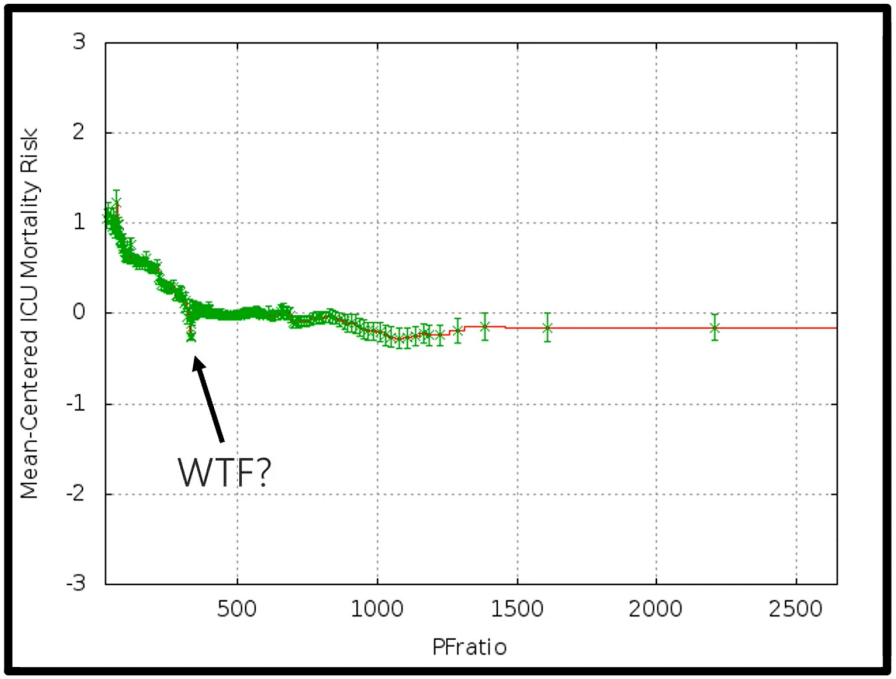
Transformations

- Most glass-box models can benefit from careful preprocessing and training.
- For example, a linear model with discretized features is more flexible than a standard linear model.
- This is, in essence, what EBM learns!

Basis expansions, feature engineering

- Adding interactions by hand
- Or with an algorithm such as RuleFit

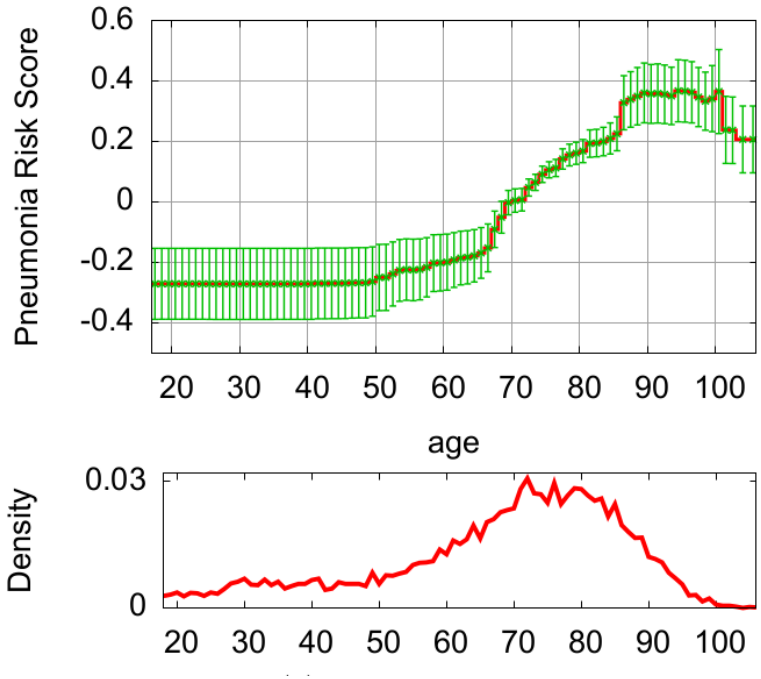
Debugging a glass-box



Source: Caruana et al.

Knowledge discovery

Age: Hypothetical explanations



Jump in risk at age 67:

- Tight error bars – jump may be real.
- Many patients in the 90's would have retired at around age 65.
- Retirement may yield differences in activity levels, health insurance, and willingness to get access to healthcare early enough to improve outcomes.

Applying domain knowledge

Constraints: e.g., Monotonicity

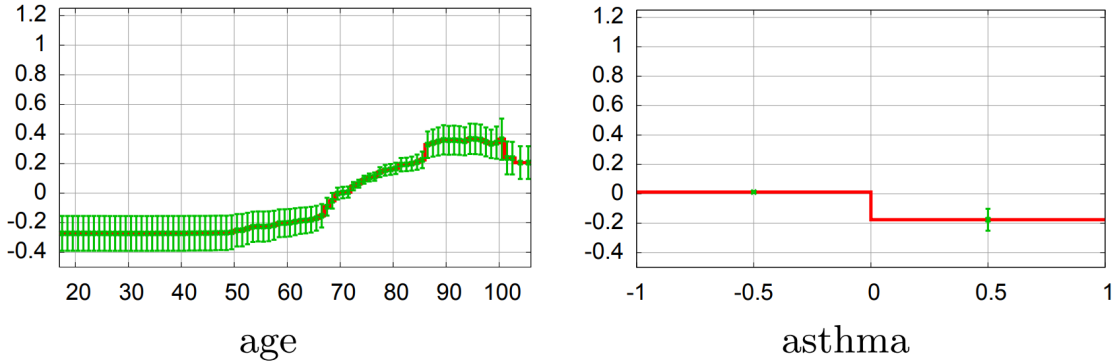
- You may want to enforce a constraint, e.g., the risk of default must decrease with the number of successful trades
- The GAM learning algorithm is unlikely to just give you back such a function
- You can enforce it

Editing the model

- It is recommended to do so post-training; during training the algorithm may compensate by using a correlated feature, potentially obscuring any monotonic violations
- Note that this will degrade the predictive performance on our data

Applying domain knowledge

Asthma



Editing the model

- Eliminate the term (making $f_j = 0$ for all x)
- Using human expertise to redraw the graph, ensuring that the risk score for asthma=1 is positive, not negative.
- See GAM Changer

Accuracy versus *correctness*

- Again, this will degrade the predictive performance on our data; however, it might make the model more *correct*.
- Applying domain knowledge is easier post- than pre-training.

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Glass-box models

- Different learning algorithms, some recent
- Usually helped by careful preprocessing
- Often account for a large fraction of the predictive power of the best-performing model; if data are very problematic, the difference might not pay off in practice.
- No conclusive evidence for a trade-off between accuracy and interpretability.
- Easier to
  - Combine with domain knowledge
  - Troubleshoot in training and production
- More suitable for high-stakes decisions

Summary

ML is not just black-boxes

- One typically aims for explaining a black-box without considering whether there is an interpretable model of similar accuracy.
- And yet, there often is!